

Chihyeon Yun (Curriculum Vitae)

AI Researcher & Engineer

1. Contact Information

- **Email:** chichi8969@naver.com
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- **Portfolio:** <https://chihyeonyoon.github.io/>
- **GitHub:** <https://github.com/ChihyeonYoon>

2. Education

- **Kyungpook National University** | Daegu, South Korea
 - **Degree:** Master of Science (M.S.) in Artificial Intelligence
 - **Period:** Mar 2023 - Feb 2025 (Graduated)
 - **GPA:** 4.11 / 4.3
 - **Laboratory:** Multimodal Language & Cognition Lab
 - **Thesis:** *Automated Multi-Channel Video Editing via Multimodal Analysis Using Deep Neural Networks*
- **Kumoh National Institute of Technology** | Gumi, South Korea
 - **Degree:** Bachelor of Science (B.S.) in Computer Engineering
 - **Period:** Mar 2020 - Feb 2023 (Graduated)
 - **GPA:** 3.16 / 4.5
- **Academic Credit Bank System** | South Korea
 - **Degree:** Associate Degree in Computer Networks
 - **Period:** Apr 2019 - Feb 2020 (Completed)
 - **GPA:** 68.0 / 100

3. Core Competencies & Technical Skills

Category	Skills & Technologies
Programming	Python (Expert), C# (Unity, Intermediate), C++ (Basic)
Deep Learning	PyTorch (Source-build capable), TensorFlow, scikit-learn
AI Research	Multimodal Late-Fusion, Audio Diffusion Models (DiffWave), VLM (Zero-shot Detection), VLA (SmolVLA), Domain Adversarial Training (DAT), Supervised Contrastive Learning (SupCon)
Audio & Vision	Librosa, SoundFile, FFmpeg, Active Speaker Detection, Pyannote.audio (Diarization), WhisperX, OpenCV, MediaPipe, Visual SLAM, 3D Gaussian Splatting (3DGS)
Infrastructure	Docker (Containerization, CUDA Env), Linux, GPU Cluster management, ROS / ROS2, Git
Methodologies	Vibe Coding (LLM-collaborative rapid prototyping), MAP Estimation, R&D Project Management (PM)

4. Research & Professional Experience

Electronics and Telecommunications Research Institute (ETRI) | Hyper-Realistic Metaverse Research Laboratory

- **Position:** Post-Master's Research Fellow
 - **Period:** Feb 2025 - Jun 2025
 - **Key Responsibilities & Achievements:**
 1. **Visual Positioning System (VPS) Development:** Developed and optimized an indoor visual localization pipeline for cyber-physical space registration, improving feature point extraction and matching under varying illumination, verified on mobile XR testbeds.
 2. **VLM-Based Semantic Monitoring:** Designed Visual-Language Model (VLM) zero-shot anomaly detection pipelines for real-time video sequence semantics on metaverse platforms.
 3. **R&D Project Management (PM):** Managed joint research milestones across multiple academic and corporate partners, inspected code quality, and authored technical specifications (Specs) and annual R&D reports.
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5. Publications

[1] SCIE Journal (Scientific Data - Nature Portfolio)

- **Title:** *A Military Audio Dataset for Situational Awareness and Surveillance* (Published, Jun 2024)
- **Authors:** June-Woo Kim, **Chihyeon Yoon (Co-author / 2nd Author)**, Ho-Young Jung
- **Contributions:** Engineered the audio preprocessing pipeline (Librosa 16kHz resampling and 80-dim log-Mel filterbank extraction) for the Military Audio Dataset (MAD, 8,075 samples) and benchmarked Audio Spectrogram Transformers (AST), achieving a peak accuracy of 91.07%.

[2] International Conference Workshop (NeurIPS 2023 DGM4H Workshop)

- **Title:** *Adversarial Fine-tuning using Generated Respiratory Sound to Address Class Imbalance* (Published, Dec 2023)
- **Authors:** June-Woo Kim, **Chihyeon Yoon (Co-author / 2nd Author)**, Miika Toikkanen, Sangmin Bae, Ho-Young Jung
- **Contributions:** Synthesized medical respiratory sound utilizing conditional audio diffusion (DiffWave) and proposed an Adversarial Fine-Tuning (AFT) framework combining a Label Classifier and Data Discriminator to align domain features, improving minority class detection by up to 26.58%.

[3] Domestic Conference & Award (KCC 2023)

- **Title:** *CutMix-SupCon: Supervised Contrastive Learning with CutMix Augmentation for Multimodal-based Emotion Recognition* (Published, Jun 2023)
- **Authors:** June-Woo Kim, Suyoung Kim, **Chihyeon Yoon (Co-author / 3rd Author)**, Jiyeon Hwang, Ho-Young Jung
- **Contributions:** Designed latent-level CutMix feature interpolation to merge KLUE-RoBERTa (text) and wav2vec 2.0 (audio) embeddings and optimized Supervised Contrastive Learning (SupCon) for cosine similarity alignment. (Awarded Bronze Prize at the 2nd ETRI Human-Understanding AI Paper Competition).

[4] SCIE Journal (Pending Submission - Lead Author)

- **Title:** *Seamless Multi-Hypothesis Video Editing Using Deep Multimodal Analysis Over Multi-Channel Sources*
- **Authors:** **Chihyeon Yun (First Author / 1st Author)**, June-Woo Kim, Soonchoul Kim, Hyeju Oh, Ho-Young Jung
- **Contributions:** Built a late-fusion frame synchronization and Beam Search decoding framework for automated multi-camera video editing, fusing WhisperX/Pyannote speaker states and Swin Transformer visual active speaker predictions, achieving a 90.10% frame alignment with expert edits.

6. Project Descriptions

[Project 1] LCA CAG 3DGS Reconstruction & Dashboard

- **Period:** Jun 2024 - Feb 2025
- **Tech Stack:** Python, PyTorch, 3D Gaussian Splatting (3DGS), OpenCV, HTML5/CSS3/JS
- **Description:** A 3D volumetric reconstruction and interactive dashboard system that combines 2D Multi-Task Deformable UNet (MT-DUNet) and Beer-Lambert radiative 3DGS to recover 3D coronary artery structure, radius profile, and volume from sparse 2D C-arm X-ray angiograms.
- **Key Responsibilities:**
 - **2D Deformable UNet (MT-DUNet):** Integrated deformable convolutions (DCNv2) to adapt to complex vessel tortuosity, predicting binary vessel masks and 25 anatomical branch labels simultaneously.
 - **Class-Constrained Triangulation:** Prevented ghost-point matching artifacts by pairing identical anatomical labels, yielding a 4.5x denser 3D centerline seed.
 - **Beer-Lambert Radiative 3DGS:** Solves the transparency occlusion problem of X-ray projection by analytically integrating density along rays, optimizing vessel radius (r) and opacity (α).
 - **Memory Optimization:** Implemented multi-resolution rendering and point downsampling, achieving an **8x GPU memory saving** (running on 2.5 GiB) on an L4 GPU.

- **Key Achievements:**
 - Achieved **76.55% Dice** on the ARCADE test set, outperforming the YOLOv8x baseline (49.00%).
 - Achieved **84.96% Mean Projection Dice** on novel-view projections.
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[Project 2] Chihyeon.ai Personal RAG Assistant & System Proxy

- **Period:** Jan 2025 - Feb 2025
 - **Tech Stack:** Python, Gemini 2.5 Flash, Pinecone Serverless, Vercel Functions, LlamaParse, Vanilla JS
 - **Description:** An interactive personal RAG system that serves as an AI secretary, providing high-fidelity, streaming answers based on academic papers, CV, and project docs.
 - **Key Responsibilities:**
 - **High-Fidelity Parsing:** Leveraged LlamaParse and Jina Reader API to extract complex academic tables, mathematical formulas, and unstructured document layouts.
 - **Secure Serverless Backend:** Created a Vercel Serverless Function proxy to secure Gemini and Pinecone credentials, preventing API key exposure on the client-side.
 - **Streaming Dialogue:** Powered multi-turn conversations with Pinecone (3072-dim) vector queries and Gemini 2.5 Flash.
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[Project 3] Unity Pick & Place Integration with SmolVLA

- **Period:** Nov 2024 - Jan 2025
 - **Tech Stack:** Python, SmolVLA, Unity Engine (C#), ROS / ROS2, Docker, PyTorch
 - **Description:** End-to-end integration of the SmolVLA (Vision-Language-Action) model within a Unity simulation environment to perform robotic 'Pick and Place' tasks.
 - **Key Responsibilities:**
 - **LLM-Collaborative Vibe Coding:** Partnered with AI code assistants to write boilerplate ROS-Unity sockets, shifting focus to SmolVLA inference and reasoning.
 - **ROS-Unity Closed Loop:** Created a real-time ROS pipeline mapping Unity visual feeds to the DL model and feedback joint torque commands back into the physics engine.
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7. Certifications, Languages & Awards

Certifications & Languages

- **Advanced Data Analytics Semi-Professional (ADsP)** | Korea Data Agency (Nov 2025)
- **Computer Literacy 1st Class** | Korea Chamber of Commerce and Industry (Nov 2019)
- **TOEIC Speaking (IM2)** | YBMTrust (Jul 2024)

Awards

- **Bronze Prize** | 2nd ETRI Human-Understanding AI Paper Competition (Jun 2023)
- **Startup Idea Award** | Graduate School of Transportation and Infrastructure Safety Startup Competition (Sep 2024)